

1. Stock Volatility & Moving Average Crossover Strategy

- **Core Concept:** Time series analysis and basic algorithmic trading logic.
- **What You'll Build:** A Python script using yfinance (financial and market data from yahoo finance) to pull historical price data for a stock (e.g., AAPL). Calculate 20-day and 50-day simple moving averages (SMA), generate buy/sell signals when the 20-day crosses above or below the 50-day, and compute simple strategy returns vs. buy-and-hold.

2. Simple "Paper Trading" Strategy Backtester

- **What it is:** A simulator that tests a basic trading rule on past stock data to see if it would have made or lost money.
- **What you learn:** Basic algorithmic logic, signal generation, and trade tracking.
- **Step-by-step tasks:**
 - Define a rule: For example, *"Buy stock whenever today's closing price is lower than yesterday's, and sell whenever today's price is higher than yesterday's."*
 - Iterate through a year of stock prices step-by-step.
 - Start with a hypothetical cash balance (e.g., \$10,000) and track trades made according to your rule.
 - Compare your ending bank account balance against a simple "buy and hold" strategy (just buying on Day 1 and sitting on it).

3. Interactive Financial Dashboard

- **What it is:** A clean web application where a user types in any stock ticker symbol and sees customized metrics.
- **What you learn:** Building user interfaces for quant tools and combining data processing with visual presentation.
- **Step-by-step tasks:**
 - Add an input box where users type a ticker symbol (e.g., NVDA, TSLA, MSFT).
 - Display key stats: current price, 52-week high/low, and annual volatility (standard deviation of daily returns).
 - Display an interactive price chart.

4. Markowitz Mean-Variance Portfolio Optimization

- **Core Concept:** Modern Portfolio Theory (MPT) and constrained numerical optimization.
- **What You'll Build:** Select 5–10 stocks, fetch historical daily returns, and calculate the covariance matrix. Use `scipy.optimize` to calculate the optimal portfolio weights that maximize the Sharpe ratio or minimize variance for a target return, then plot the **Efficient Frontier**.

5. Pairs Trading Strategy (Statistical Arbitrage)

- **Core Concept:** Cointegration and mean-reversion trading.
- **What You'll Build:** Identify two historically correlated stocks (e.g., KO and PEP). Perform a cointegration test (`statsmodels.tsa.stattools.coint`) to verify their price spread is stationary. Track the spread's z-score and simulate trades when the spread deviates more than 2 standard deviations from its historical mean.